

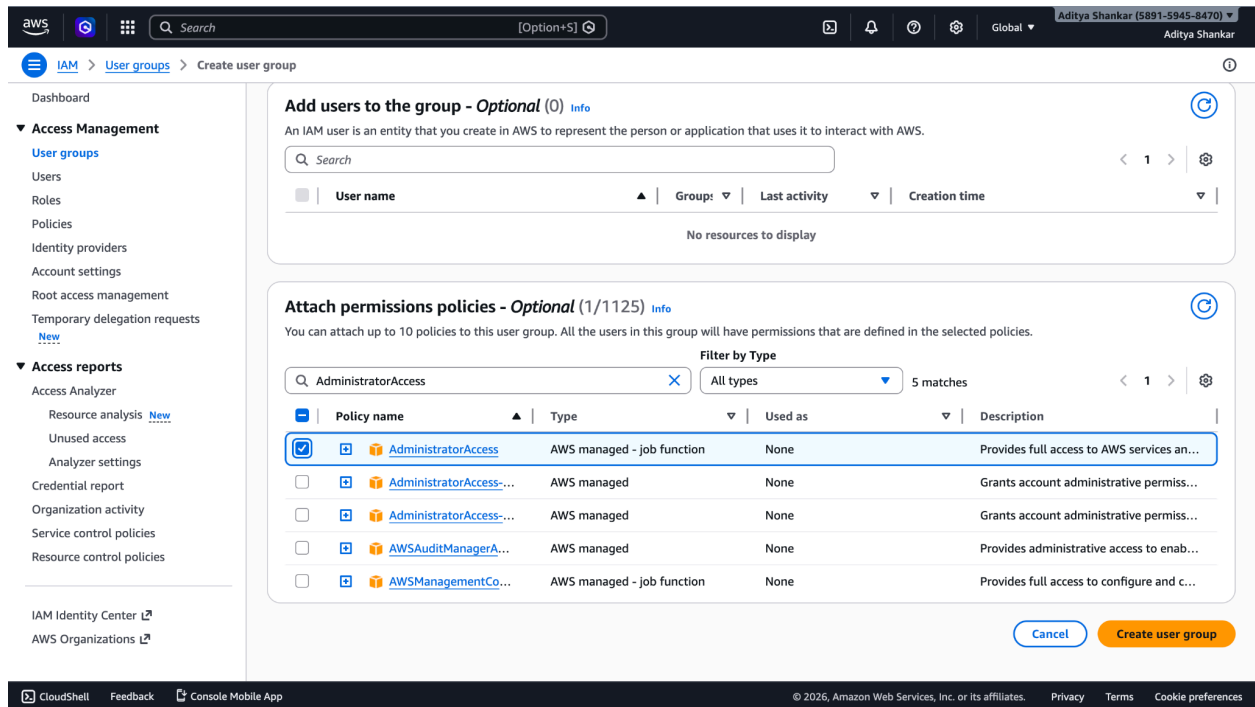
AWS Hands-On Lab IAM + VPC + EC2

Date: 24th March 2026

Account: Aditya Shankar (5891-5945-8470)

IAM User used: shankdevops

Lab 1 IAM (Identity and Access Management)



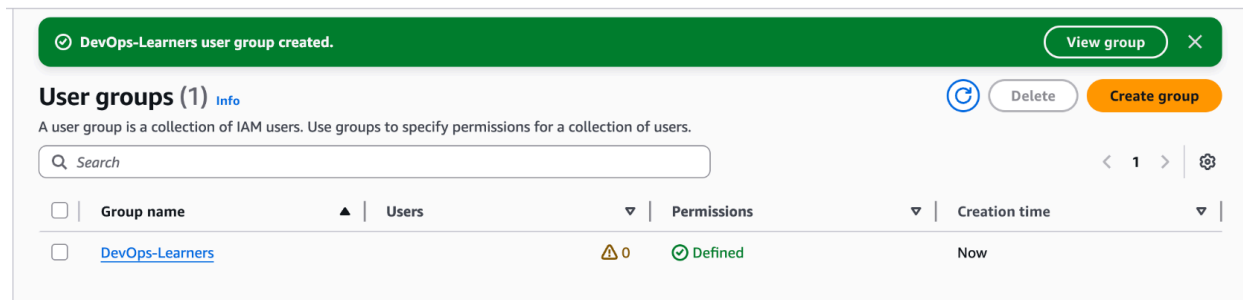
The screenshot shows the AWS IAM console interface. The left sidebar contains navigation links for Dashboard, Access Management (User groups, Users, Roles, Policies, Identity providers, Account settings, Root access management, Temporary delegation requests), Access reports (Access Analyzer, Resource analysis, Unused access, Analyzer settings, Credential report, Organization activity, Service control policies, Resource control policies), IAM Identity Center, and AWS Organizations. The main content area is titled 'Create user group'. It has two sections: 'Add users to the group - Optional (0)' and 'Attach permissions policies - Optional (1/1125)'. The first section is empty. The second section shows a search for 'AdministratorAccess' with 5 matches. The first match, 'AdministratorAccess', is selected. The table below shows the details of the policies.

Policy name	Type	Used as	Description
☒ AdministratorAccess	AWS managed - job function	None	Provides full access to AWS services an...
☐ AdministratorAccess...	AWS managed	None	Grants account administrative permis...
☐ AdministratorAccess...	AWS managed	None	Grants account administrative permis...
☐ AWSAuditManagerA...	AWS managed	None	Provides administrative access to enab...
☐ AWSManagementCo...	AWS managed - job function	None	Provides full access to configure and c...

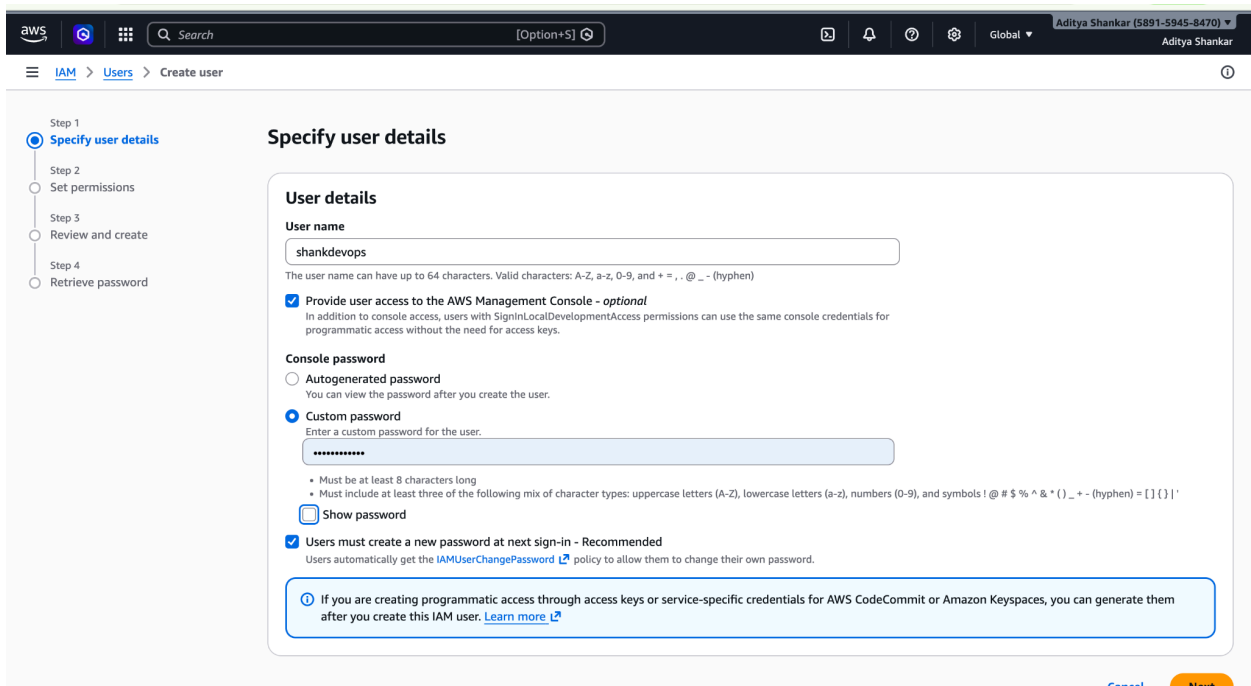
At the bottom right of the 'Attach permissions policies' section, there are two buttons: 'Cancel' and 'Create user group'.

What I did

Step 1 User Group creation Created an IAM User Group named **DevOps-Learners** and attached the **AdministratorAccess** AWS managed policy to it. In production this would be scoped down to least privilege. **AdministratorAccess** is used here because this is a personal learning account with no sensitive workloads.



Step 2 IAM User creation Created an IAM user **shankdevops** with AWS Management Console access enabled and a custom password. Added the user to the **DevOps-Learners** group so it inherits the group's permissions rather than having policies attached directly this is the correct AWS best practice (manage permissions at group level, not user level).



Step 3 Switched to IAM user Signed in to the AWS Console as **shankdevops** using the account-specific sign-in URL (<https://589159458470.signin.aws.amazon.com/console>). All subsequent work was done as this IAM user, not root. This is the correct real-world habit root is only used for account-level tasks like billing.

IAM > Users > Create user

Step 1

Specify user details

Step 2

Set permissions

Step 3

Review and create

Step 4

Retrieve password

Set permissions

Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. [Learn more](#)

Permissions options

☒ **Add user to group**
Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ **Copy permissions**
Copy all group memberships, attached managed policies, and inline policies from an existing user.

☐ **Attach policies directly**
Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

User groups (1)

☐

Group name

▲

Users

▼

Attached policies

▼

Created

☐

DevOps-Learners

0

[AdministratorAccess](#)

2026-03-24 (5 minutes ago)

Set permissions boundary - optional

Cancel

Previous

Next

IAM > Users > Create user

Step 1

Specify user details

Step 2

Set permissions

Review and create

Step 4

Retrieve password

Review and create

Review your choices. After you create the user, you can view and download the autogenerated password, if enabled.

User details

User name
shankdevops

Console password type
Custom password

Require password reset
Yes

Permissions summary

▲

Type

▼

Used as

▼

[IAMUserChangePassword](#)

AWS managed

Permissions policy

Tags - optional

Tags are key-value pairs you can add to AWS resources to help identify, organize, or search for resources. Choose any tags you want to associate with this user.

No tags associated with the resource.

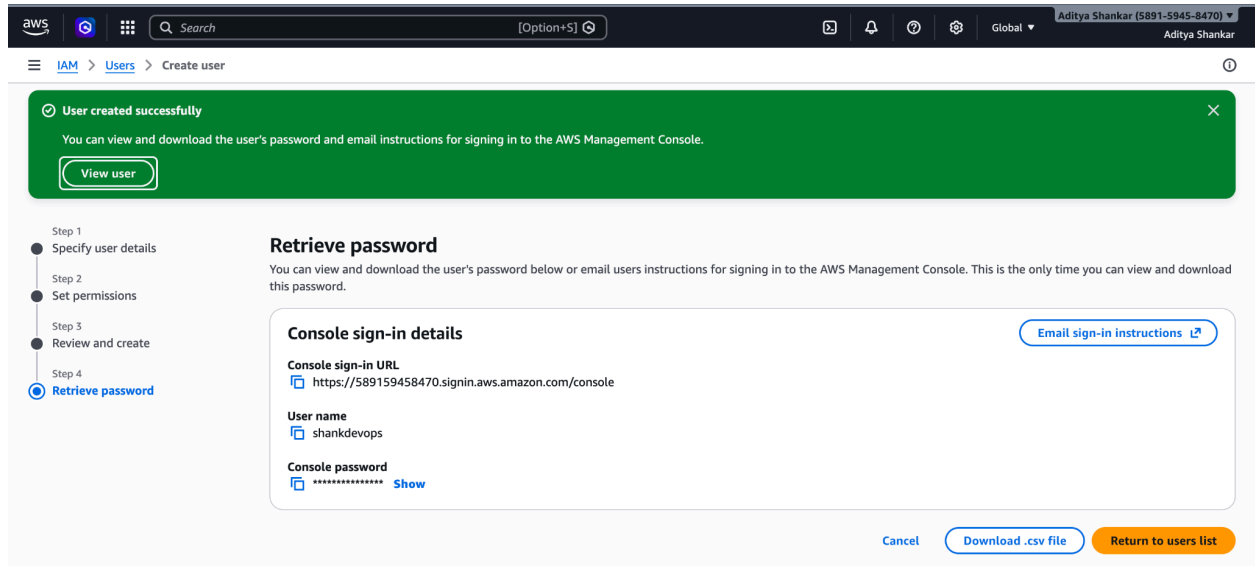
Add new tag

You can add up to 50 more tags.

Cancel

Previous

Create user



Verified identity by expanding the top-right account menu which showed:

- Account name: Aditya Shankar
- IAM user: shankdevops

Key concepts understood

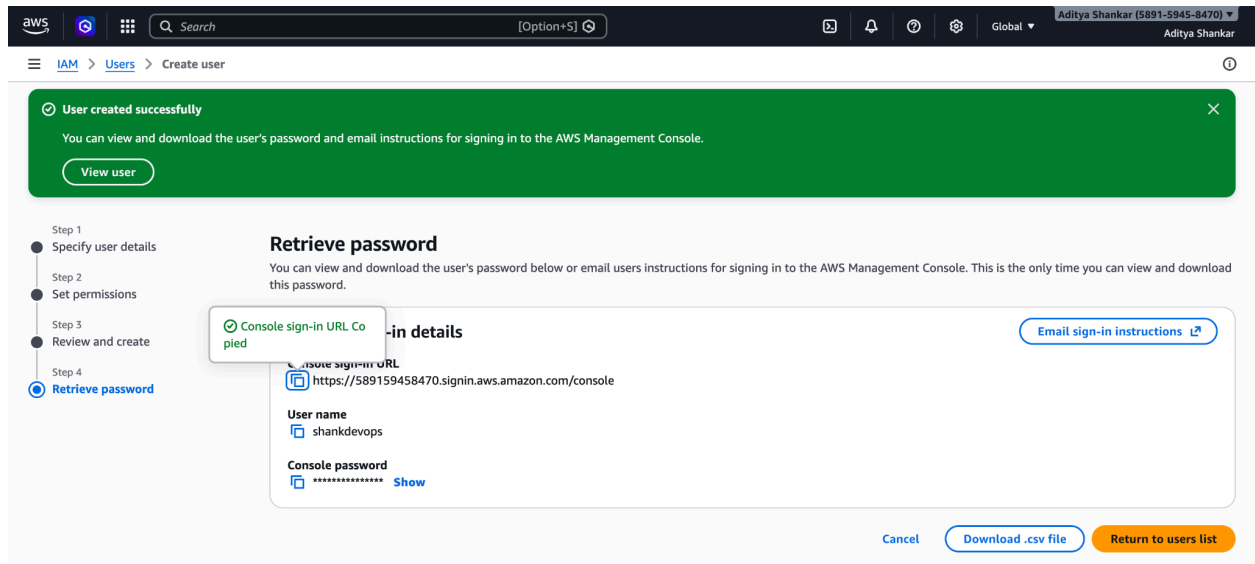
- IAM is **global** not tied to any region
- **Users** represent people; **Groups** are collections of users; **Policies** are JSON permission documents
- Permissions flow from Policy → Group → User, not the other way around
- The sign-in URL is account-specific and embeds the 12-digit account ID
- Root account should never be used for day-to-day work

Lab 2 VPC (Virtual Private Cloud)

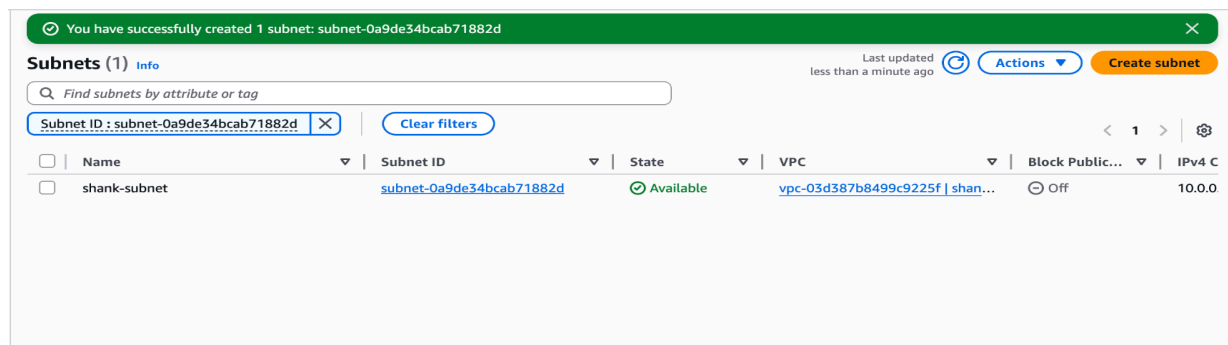
What I did

Step 1 Created a custom VPC Created a VPC named **shank-vpc** with CIDR block **10.0.0.0/16** in the us-east-1 (N. Virginia) region. This gives 65,536 available IP addresses within the private network.

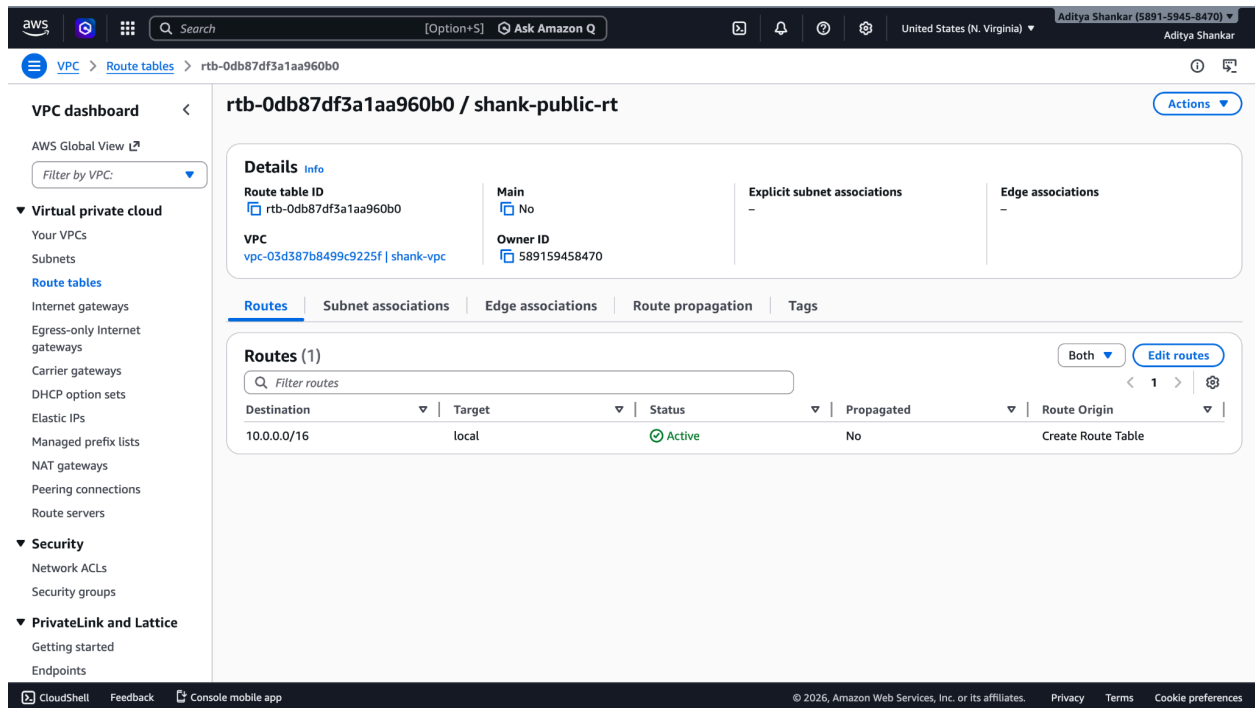
Chose not to use the default VPC that AWS provides in every region real projects always use custom VPCs for isolation and control.



Step 2 Created a public subnet Created **shank-subnet** inside **shank-vpc**. A subnet is a subdivision of the VPC's IP range placed in a specific Availability Zone. This subnet was given CIDR **10.0.0.0/24**, providing 256 addresses within it.



Step 3 Route table Examined the automatically created route table **shank-public-rt** (rt-0db87df3a1aa960b0) associated with **shank-vpc**. The default route **10.0.0.0/16** → **local** was already present, which allows all resources inside the VPC to communicate with each other. To make a subnet truly public, a route **0.0.0.0/0** → **Internet Gateway** needs to be added this is the next step for when an IGW is attached.



Step 4 Security Group Created Security Group `shank-web-sg` inside `shank-vpc` with the following inbound rules:

Type	Protocol	Port	Source
SSH	TCP	22	My IP only
HTTP	TCP	80	0.0.0.0/0 (anywhere)

SSH is restricted to My IP only as a security measure exposing port 22 to the world is a common misconfiguration that leads to brute-force attacks. HTTP is open to the world because this simulates a web server that should be publicly accessible.

Security group (sg-003a416d92587701d | shank-web-sg) was created successfully
Details

sg-003a416d92587701d - shank-web-sg

Actions

Details

Security group name
shank-web-sg

Security group ID
sg-003a416d92587701d

Description
Allow SSH and HTTP inbound

VPC ID
vpc-02b2df7cf245f4ce5

Owner
589159458470

Inbound rules count
2 Permission entries

Outbound rules count
1 Permission entry

Inbound rules
Outbound rules
Sharing
VPC associations
Related resources - new
Tags

Inbound rules (2)

Manage tags
Edit inbound rules

Search

☐
Name

☐
Security group rule ID

☐
IP version

☐
Type

☐
Protocol

☐
Port range

☐
-

sgr-036561b3929e1dc8a

IPv4

SSH

TCP

22

☐
-

sgr-04ad0286f02d422a6

IPv4

HTTP

TCP

80

Outbound: all traffic allowed (default) instances can reach the internet freely.

Step 5 EC2 instance launched Launched an EC2 instance inside `shank-vpc` using the custom security group `shank-web-sg`. Instance ID: `i-0d05af7d426c2268f`. The instance launch success page confirmed the instance was placed in the custom VPC, not the default one.

aws

Search

[Option+S]

United States (N. Virginia)

Aditya Shankar (5891-5945-8470)

EC2

Instances

Launch an instance

Access

Successfully initiated launch of instance (i-0d05af7d426c2268f)

Launch log

Next Steps

What would you like to do next with this instance, for example "create alarm" or "create backup"

1

2

3

4

5

6

Create billing usage alerts

To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.

Create billing alerts

Connect to your instance

Once your instance is running, log into it from your local computer.

Connect to instance

Learn more

Connect an RDS database

Configure the connection between an EC2 instance and a database to allow traffic flow between them.

Connect an RDS database

Create a new RDS database

Learn more

Create EBS snapshot policy

Create a policy that automates the creation, retention, and deletion of EBS snapshots

Create EBS snapshot policy

Manage detailed monitoring

Enable or disable detailed monitoring for the instance. If you

Create Load Balancer

Create a application, network gateway or classic Elastic Load Balancer

Create AWS budget

AWS Budgets allows you to create budgets, forecast spend, and take action on your costs and usage

Manage CloudWatch alarms

Create or update Amazon CloudWatch alarms for the

Info

Tutorial

Launch a test instance - Post Launch Setup

Task 1: Post launch setup

Total steps: 2

Task 2: Terminate (delete) your instance

Total steps: 2

Step 1: Navigate to the instance details page

Step 2: Change instance state to terminate (delete)

Dismiss tutorial

Key concepts understood

- A VPC is a logically isolated private network inside AWS other AWS customers cannot see or reach it

- CIDR block **10.0.0.0/16** means the first 16 bits are fixed giving a range of **10.0.0.0** to **10.0.255.255**
- A subnet is always tied to one Availability Zone for high availability, you spread subnets across multiple AZs
- What makes a subnet "public" is a route table entry pointing **0.0.0.0/0** to an Internet Gateway not any property of the subnet itself
- Security Groups are **stateful** if you allow inbound SSH, the return traffic is automatically allowed without an explicit outbound rule
- Security Groups are the firewall at the instance level; Network ACLs operate at the subnet level (stateless)

VPC > Security Groups > sg-003a416d92587701d - shank-web-sg

Security group (sg-003a416d92587701d | shank-web-sg) was created successfully

sg-003a416d92587701d - shank-web-sg

Details

Security group name shank-web-sg	Security group ID sg-003a416d92587701d	Description Allow SSH and HTTP inbound	VPC ID vpc-02b2df7cf245f4ce5
Owner 589159458470	Inbound rules count 2 Permission entries	Outbound rules count 1 Permission entry	

Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - new | Tags

Inbound rules (2)

Name	Security group rule ID	IP version	Type	Protocol	Port range
-	sgr-036561b3929e1dc8a	IPv4	SSH	TCP	22
-	sgr-04ad0286f02d422a6	IPv4	HTTP	TCP	80

IAM > User groups > Create user group

Add users to the group - Optional (0)

An IAM user is an entity that you create in AWS to represent the person or application that uses it to interact with AWS.

Attach permissions policies - Optional (1/1125)

You can attach up to 10 policies to this user group. All the users in this group will have permissions that are defined in the selected policies.

Filter by Type

AdministratorAccess 5 matches

Policy name	Type	Used as	Description
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☐ AdministratorAccess...	AWS managed	None	Grants account administrative permis...
☐ AdministratorAccess...	AWS managed	None	Grants account administrative permis...
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☐ AWSManagementCo...	AWS managed - job function	None	Provides full access to configure and c...